

Ex.no:4	EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION

Aim:

To perform data inspection, conditional filtering, sorting, statistical summary, group aggregation, correlation analysis, and data visualization on a hotel bookings dataset using Pandas and Matplotlib.

Algorithm:

1. Start.
2. Import the required Python libraries: pandas and matplotlib.pyplot.
3. Load the dataset hotel_bookings.csv into a pandas DataFrame using pd.read_csv().
4. Inspect the basic structure, initial records, shape, column data types, and numerical summaries using df.head(), df.shape, df.dtypes, and df.describe().
5. Filter rows based on specific business conditions (e.g., adr > 100 and hotel == "Resort Hotel").
6. Sort the dataset by adr in descending order using df.sort_values().
7. Compute statistical metrics for adr including mean, median, mode, and standard deviation.
8. Group data by hotel type and aggregate to find the average adr using df.groupby().
9. Calculate the pairwise correlation matrix for numerical features using df.corr().
10. Generate a bar plot for average adr by hotel type and a scatter plot of lead_time versus adr using Matplotlib.

Program:

```
import pandas as pd

import matplotlib.pyplot as plt

# Import Dataset

df = pd.read_csv("/content/hotel_bookings.csv")

# Dataset Inspection

print("===== FIRST FIVE RECORDS =====")

display(df.head())

print("\n===== SHAPE OF DATASET =====")

print(df.shape)

print("\n===== DATA TYPES =====")

print(df.dtypes)

print("\n===== SUMMARY =====")

display(df.describe())

# Filtering Data

# Bookings with ADR greater than 100

print("\n===== BOOKINGS WITH ADR > 100 =====")

high_adr = df[df["adr"] > 100]

display(high_adr.head())

# Resort Hotel bookings

print("\n===== RESORT HOTEL BOOKINGS =====")

resort = df[df["hotel"] == "Resort Hotel"]

display(resort.head())

# Sorting Data

print("\n===== SORT BY ADR (DESCENDING) =====")
```

```

sorted_df = df.sort_values(by="adr", ascending=False)

display(sorted_df.head())

# Statistical Analysis

print("\n===== ADR STATISTICS =====")

print("Mean :", df["adr"].mean())

print("Median :", df["adr"].median())

print("Mode :")

print(df["adr"].mode())

print("Standard Deviation :", df["adr"].std())

print("\n===== DESCRIPTIVE STATISTICS =====")

display(df.describe())

# Grouping and Aggregation

print("\n===== AVERAGE ADR BY HOTEL TYPE =====")

hotel_avg = df.groupby("hotel")["adr"].mean()

display(hotel_avg)

# Correlation Analysis

print("\n===== CORRELATION =====")

corr = df[["lead_time", "stays_in_weekend_nights", "stays_in_week_nights", "adults",
"children", "babies", "adr", "is_canceled"]].corr()

display(corr)

# Visualization 1

hotel_avg.plot(kind="bar")

plt.title("Average ADR by Hotel Type")

plt.xlabel("Hotel")

plt.ylabel("Average ADR")

plt.show()

```

```
# Visualization 2

plt.figure(figsize=(8,5))

plt.scatter(df["lead_time"], df["adr"])

plt.title("Lead Time vs ADR")

plt.xlabel("Lead Time")

plt.ylabel("ADR")

plt.show()
```

Output:

----- FIRST FIVE RECORDS -----

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

----- SHAPE OF DATASET -----
(112457, 8)

----- DATA TYPES -----

Index object
Date object
Open float64
High float64
Low float64
Close float64
Adj Close float64
Volume float64
dtype: object

----- SUMMARY -----

	Open	High	Low	Close	Adj Close	Volume
count	110253.000000	110253.000000	110253.000000	110253.000000	110253.000000	1.102530e+05
mean	7658.515222	7704.372961	7608.000422	7657.545872	7657.351729	1.273975e+09
std	9011.478913	9066.638548	8954.506981	9011.510444	9011.608900	4.315783e+09
min	54.869999	54.869999	54.869999	54.869999	54.869999	0.000000e+00
25%	1855.030029	1864.510010	1843.979980	1855.060059	1854.179565	0.000000e+00
50%	5194.089844	5226.279785	5154.049805	5194.750000	5194.750000	4.329000e+05
75%	10134.299810	10207.820310	10060.349610	10134.830080	10134.830080	1.734314e+08
max	68775.062500	69403.750000	68516.992190	68775.062500	68775.062500	9.440374e+10

----- RECORDS WITH CLOSE > 100 -----

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

----- RECORDS WITH HIGH VOLUME -----

	Index	Date	Open	High	Low	Close	Adj Close	Volume
8815	NYA	2001-01-03	6968.830078	6968.830078	6968.830078	6968.830078	6968.830078	1.880700e+09
8816	NYA	2001-01-04	6894.810059	6894.810059	6894.810059	6894.810059	6894.810059	2.169400e+09
8817	NYA	2001-01-05	6774.060059	6774.060059	6774.060059	6774.060059	6774.060059	1.430800e+09
8818	NYA	2001-01-08	6759.049805	6759.049805	6759.049805	6759.049805	6759.049805	1.115500e+09
8819	NYA	2001-01-09	6754.819824	6754.819824	6754.819824	6754.819824	6754.819824	1.191300e+09

----- SORT BY CLOSE (DESCENDING) -----

	Index	Date	Open	High	Low	Close	Adj Close	Volume
92396	J203.JO	2021-03-11	68516.99219	68924.99219	68516.99219	68775.06250	68775.06250	0.0
92419	J203.JO	2021-04-16	67823.09375	68741.82813	67823.09375	68698.78125	68698.78125	0.0
92394	J203.JO	2021-03-09	68426.17188	69021.59375	68051.92188	68655.96094	68655.96094	0.0
92433	J203.JO	2021-05-07	67608.89063	68664.35938	67608.89063	68519.53125	68519.53125	0.0
92395	J203.JO	2021-03-10	68655.96094	68655.96094	67941.45313	68516.99219	68516.99219	0.0

----- CLOSE PRICE STATISTICS -----
Mean : 7657.545871842671
Median : 5194.75
Mode :
0 3187.628117
Name: Close, dtype: float64
Standard Deviation : 9011.518443538464

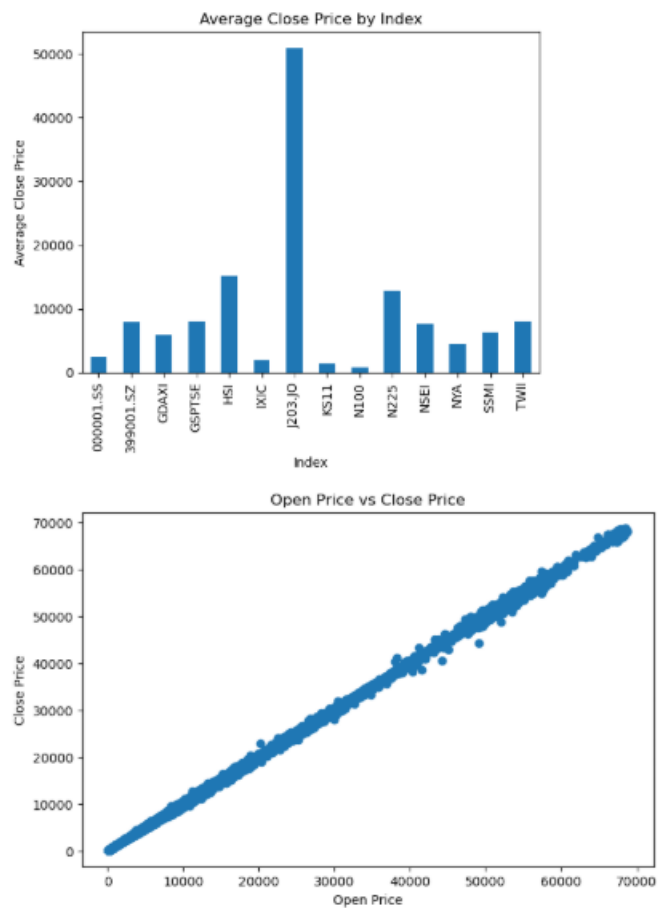
----- DESCRIPTIVE STATISTICS -----

	Open	High	Low	Close	Adj Close	Volume
count	110253.000000	110253.000000	110253.000000	110253.000000	110253.000000	1.102530e+05
mean	7658.515222	7704.372961	7608.000422	7657.545872	7657.351729	1.273975e+09
std	9011.478913	9066.638548	8954.506981	9011.510444	9011.608900	4.315783e+09
min	54.869999	54.869999	54.869999	54.869999	54.869999	0.000000e+00
25%	1855.030029	1864.510010	1843.979980	1855.060059	1854.179565	0.000000e+00
50%	5194.089844	5226.279785	5154.049805	5194.750000	5194.750000	4.329000e+05
75%	10134.299810	10207.820310	10060.349610	10134.830080	10134.830080	1.734314e+08
max	68775.062500	69403.750000	68516.992190	68775.062500	68775.062500	9.440374e+10

----- AVERAGE CLOSE BY INDEX -----
Index
000001.SS 2383.869135
399001.SZ 7973.831805
GDAXI 5914.846717
S5P1SE 8090.866305
HSI 15200.605629
IXIC 1984.986795
J203.JO 58910.945859
K511 1489.145147
N100 822.153719
N225 12849.155197
NSEI 7668.847238
NYA 4452.174711
SSMI 6489.733878
TWII 8028.866141
Name: Close, dtype: float64

----- CORRELATION -----

	Open	High	Low	Close	Adj Close	Volume
Open	1.000000	0.999959	0.999944	0.999907	0.999907	0.138673
High	0.999959	1.000000	0.999931	0.999955	0.999955	0.138510
Low	0.999944	0.999931	1.000000	0.999960	0.999960	0.138549
Close	0.999907	0.999955	0.999960	1.000000	1.000000	0.138555
Adj Close	0.999907	0.999955	0.999960	1.000000	1.000000	0.138560
Volume	0.138673	0.138510	0.138549	0.138555	0.138560	1.000000



Result: